

GDSQ-VLA: Geometry- and Distribution-Sensitive Layer Selection for Post-Training Quantization of Vision-Language-Action Models

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Abstract

Post-training quantization can reduce the deployment storage of vision-language-action (VLA) policies, yet uniform low precision is brittle because language reasoning and diffusion-based action generation exhibit strongly heterogeneous layer sensitivity. We present GDSQ-VLA, a training-free layer-selection framework that treats quantization robustness as both a representation-geometry and a distribution-overlap problem. For each candidate Linear layer, GDSQ-VLA compares an isolated W4A8 intervention with a full-precision reference using centered kernel alignment (CKA) and Cauchy-Schwarz (CS) divergence, weights the resulting score by paired action-generation sensitivity, and rejects infeasible choices with activation-magnitude and saturation guards. A budgeted selector then proposes W4/FP16 masks, while paired functional adjudication captures cross-layer error accumulation that additive proxies miss. On RoboCasa365 with GR00T N1.5, a predeclared 200-scenario development study selects a CKA:CS ratio of 16:1. On 14 atomic tasks excluded from this selection, GDSQ-VLA reaches 65.9% success, compared with 45.6% for the original W4A8 QuantVLA recipe, a 20.3 percentage-point gain with a 95% task-cluster interval of [13.9, 26.9]. The resulting model quantizes 100 of 180 LLM+DiT Linear layers and provides a theoretical $1.99\times$ component-storage compression, while remaining 8.0 points below FP16. These results show that dual-similarity layer allocation recovers a substantial fraction of low-bit control quality, but does not eliminate the accuracy-compression trade-off.

1. Introduction

Vision-language-action (VLA) models connect visual perception and language-conditioned reasoning to continuous robot control, enabling one policy to execute diverse instructions across objects, scenes, and embodiments [3, 4, 9, 13, 26]. Diffusion- and flow-based policies further provide expressive continuous action generation [2, 6, 21, 25, 27]. Their

capabilities, however, come with a deployment cost: large language backbones and generative action heads require substantial weight storage and memory bandwidth, while control loops impose strict latency and hardware constraints. Post-training quantization (PTQ) is attractive in this setting because it changes numerical precision without retraining the policy or redesigning its architecture [19, 36].

The central challenge is that a VLA policy is not uniformly tolerant to quantization. Its language backbone produces a multimodal conditioning representation, and its action head repeatedly transforms a noise sample into an executable action chunk. A small representation error can therefore alter both the geometry of task-conditioned features and their distribution, then accumulate through the action solver. Uniform W4A8 quantization ignores this heterogeneity. Fixed architectural layouts, including the selective attention protection and scale calibration used by QuantVLA [40], protect known fragile interfaces but cannot decide which individual language and action layers should remain in full precision for a new policy checkpoint.

Existing mixed-precision methods commonly rank layers using local reconstruction error, activation magnitude, Hessian sensitivity, or task-specific proxy metrics [5, 7, 8, 10, 12, 42]. These criteria provide useful signals, but each exposes only one view of representation damage. Centered kernel alignment (CKA) detects whether relationships among samples are preserved [14, 23, 28], yet is intentionally invariant to isotropic scaling. Cauchy-Schwarz (CS) divergence instead measures distribution overlap [39], but depends on a kernel bandwidth and may be noisy across layers. Neither metric alone indicates how strongly a representation change affects downstream robot actions.

Our insight is that reliable VLA precision allocation requires complementary representation evidence and a behavioral referee. Geometry and distribution should be measured under the same isolated intervention protocol, because their disagreement identifies failure modes that a single metric suppresses. Their layer score should then be scaled by an action-level sensitivity measured with paired solver noise. Fi-

nally, feasibility guards and configuration-level action comparison are needed because representation similarity is not a guarantee of stable activation range or compositional behavior.

We instantiate this insight as GDSQ-VLA, a geometry- and distribution-sensitive layer-selection framework for W4/FP16 PTQ. We probe each of 116 candidate Linear layers by applying W4A8 to that layer while the remaining candidate layers follow a full-precision attribution path. The probe computes linear CKA and CS divergence, combines their normalized distortions with an action-derived weight, and removes layers that violate activation RMS or estimated saturation thresholds. Under a uniform-W6-equivalent storage budget, a selector generates diverse masks and a paired functional score adjudicates the most promising complete configurations. Closed-loop development evaluation freezes the final CKA:CS ratio at 16:1 and produces a mask with 100 W4A8 candidate layers and 16 FP16 skips; 64 DiT attention projections remain protected outside the candidate set.

We evaluate the frozen method on RoboCasa365 [24], using official task horizons, four denoising steps, 16 executed actions per replan, and common random numbers for paired policy comparison. The primary result excludes the four tasks used for ratio selection. Across the remaining 14 atomic tasks and 700 episodes per configuration, GDSQ-VLA obtains 65.9% success versus 45.6% for QuantVLA W4A8 with static ATM/OHB. The paired task-level improvement is 20.3 points with a 95% interval of [13.9, 26.9]. GDSQ-VLA retains a theoretical $1.99\times$ LLM+DiT component-storage compression, although its current eager fake-quant implementation neither accelerates inference nor reduces live GPU memory.

Our contributions are threefold:

- (1) We formulate VLA layer selection as action-weighted geometry and distribution preservation, combining CKA and CS divergence with explicit magnitude and saturation feasibility guards.
- (2) We introduce a two-level selection procedure that uses additive layer proxies for efficient search and paired functional adjudication for cross-layer action error, yielding a reproducible W4/FP16 mask under a fixed storage budget.
- (3) We provide a controlled RoboCasa365 evaluation that separates development and primary tasks, reports paired uncertainty and ablations, and states the deployment-storage benefit separately from the limitations of the current fake-quant runtime.

2. Related Work

Vision-language-action models. Generalist robot policies have progressed from large-scale transformer control and embodied language models [3, 4, 9] to reusable multi-robot and open VLA policies [13, 26]. Continuous-control

systems increasingly use diffusion or flow-based action generation: Diffusion Policy established action-space denoising [6], while RDT-1B, π_0 , $\pi_{0.5}$, and GR00T combine large multimodal encoders with generative action heads [2, 21, 25, 27]. This coupling improves generality but makes numerical errors consequential both in semantic conditioning and across iterative action refinement.

Efficient VLA inference. Efficiency work reduces VLA cost through smaller policies, dynamic depth, token reuse, and cache reuse. SmolVLA targets an affordable compact model [31]; MoLe-VLA routes examples through selected language layers [41]; EfficientVLA combines language-layer pruning, visual-token reduction, and action-head caching [38]; and VLA-Cache reuses temporally stable token features [37]. These approaches are complementary to quantization, but they change training, execution, or token flow. Our setting leaves the checkpoint and operator graph intact and allocates numerical precision after training.

Transformer and multimodal PTQ. Weight-only PTQ methods such as GPTQ and AWQ optimize rounding or protect activation-salient channels [10, 20]. Weight-activation methods instead reshape difficult activation distributions: SmoothQuant migrates channel outliers into weights [36], OmniQuant learns calibration transformations [30], and rotation-based methods redistribute outliers before quantization [1, 11, 19, 22, 32]. Block reconstruction provides another calibration objective [18]. Multimodal PTQ further accounts for modality imbalance in vision-language models through calibrated scaling or modality-aware objectives [16, 33]. These methods improve the low-bit substrate, whereas our primary question is which individual VLA layers should use that substrate.

Diffusion and VLA quantization. Diffusion PTQ must accommodate timestep-dependent activation statistics and error accumulation. PTQ4DM and Q-Diffusion introduced timestep-aware calibration for convolutional diffusion models [17, 29]; PTQ4DiT, Q-DiT, and ViDiT-Q extend quantization to transformer generators [5, 35, 43]; MixDQ searches metric-decoupled mixed precision [42]; and SVDQuant isolates outliers in low-rank branches [15]. QuantVLA is the closest VLA-specific PTQ baseline: it applies DuQuant-style transformations to the language and action stacks while calibrating attention temperature and output-head energy [40]. GDSQ-VLA uses the same W4A8 substrate for controlled comparison, but replaces a fixed protection layout with checkpoint-specific layer evidence and budgeted W4/FP16 allocation.

Representation similarity and distribution alignment. SVCCA and projection-weighted CCA compare learned

representation subspaces [23, 28], while centered kernel alignment (CKA) compares sample geometry with invariance to orthogonal transformations and isotropic scaling [14]. CS divergence instead estimates distribution overlap from within-set and cross-set kernel similarities and has recently been used for distributional vision-language alignment [39]. We use CKA and CS divergence as complementary frozen probes, not training losses: they rank isolated quantization interventions, while action weighting and complete-configuration adjudication determine deployment.

Mixed-precision allocation. Mixed-precision search has been driven by reinforcement learning and hardware feedback [34], Hessian sensitivity [7, 8], reconstruction loss [12, 18], and task-specific diffusion metrics [5, 42]. Additive layer objectives are efficient but miss cancellation and amplification among simultaneous interventions; direct task search captures those interactions but scales poorly. GDSQ-VLA bridges the two by using dual-similarity evidence to generate feasible masks and reserving paired action-functional evaluation for a diverse Top- K set.

3. Method

Overview. GDSQ-VLA allocates W4A8 and FP16 at layer granularity without updating pretrained parameters. The method has four stages, illustrated in Fig. 1: (i) an isolated intervention protocol measures the effect of quantizing one layer; (ii) dual-similarity scores describe representation geometry and distribution shift; (iii) an action-weighted constrained selector proposes diverse precision masks; and (iv) paired functional adjudication chooses complete configurations before a closed-loop development set freezes the metric ratio. This organization separates inexpensive layer attribution from the cross-layer behavior that ultimately determines control quality.

3.1. Quantization Setting and Dual References

We consider a VLA policy with a vision-language backbone and a diffusion-transformer (DiT) action head. Given visual observations I , language instruction l , robot state s , and diffusion time t , the policy conditions an iterative action update

$$x_{t-1} = f_{\theta}(x_t, F_{VL}(I, l), s, t), \quad (1)$$

where the final state is decoded into an action chunk. We apply group-wise DuQuant-style W4 weights and static A8 activations with group size 64 [19]. The 116 selectable Linear layers comprise 84 language-model projections and 32 DiT feed-forward projections. The 64 DiT attention projections are outside the search space and remain FP16, following their known sensitivity to upstream scale drift [40].

A single reference cannot cleanly distinguish quantization attribution from deployment feasibility, so we use two. The

attribution reference R retains the quantization wrappers, transformations, and A8 path but sets all candidate weight bits to zero, activating the full-precision weight branch. Comparing an isolated intervention against R attributes the difference to the selected W4 weight choice rather than wrapper mechanics. The *deployment reference* is the unwrapped FP16 model. We use R for CKA, CS, and single-layer action sensitivity, while magnitude guards and complete-configuration adjudication use the deployment reference.

For candidate layer i , all candidate layers follow R except i , whose weight is set to W4. A fixed calibration buffer and paired initial action noise are reused across interventions. Hooks collect matched reference and quantized outputs (H_i^R, H_i^Q) at identical token positions. Padding rows are masked from the reference and the same row mask is applied to the quantized output; deterministic stratified pooling limits visual, language, and state tokens so that one modality does not dominate the score.

3.2. Geometry and Distribution Sensitivity

CKA measures whether quantization preserves relationships among token representations. For centered matrices $\tilde{H}_i^R, \tilde{H}_i^Q \in \mathbb{R}^{N \times d}$, we use linear CKA

$$\text{CKA}_i = \frac{\|(\tilde{H}_i^R)^\top \tilde{H}_i^Q\|_F^2}{\|(\tilde{H}_i^R)^\top \tilde{H}_i^R\|_F \|(\tilde{H}_i^Q)^\top \tilde{H}_i^Q\|_F + \epsilon}. \quad (2)$$

We define the geometric distortion as $D_i^{\text{CKA}} = 1 - \text{CKA}_i$. This score is invariant to an orthogonal basis change and global isotropic scale, which makes it a stable geometric statistic but also creates a scale blind spot.

CS divergence complements this blind spot by comparing distribution overlap. With Gaussian kernel $k(a, b) = \exp(-\|a - b\|_2^2 / (2\sigma_i^2))$ and bandwidth σ_i fixed from the reference samples, the empirical divergence is

$$D_i^{\text{CS}} = \log \left(\frac{1}{N^2} \sum_{a,b} k(h_a^R, h_b^R) \right) + \log \left(\frac{1}{N^2} \sum_{a,b} k(h_a^Q, h_b^Q) \right) - 2 \log \left(\frac{1}{N^2} \sum_{a,b} k(h_a^R, h_b^Q) \right). \quad (3)$$

We evaluate Eq. 3 with log-sum-exp accumulation and a bandwidth floor of 10^{-3} . Because each layer has its own feature distribution and bandwidth, D_i^{CS} is a ranking proxy rather than an absolute cross-layer distance.

We jointly min-max normalize D_i^{CKA} and D_i^{CS} across all candidate measurements and form

$$S_i = \lambda_{\text{CKA}} \overline{D^{\text{CKA}}}_i + \lambda_{\text{CS}} \overline{D^{\text{CS}}}_i. \quad (4)$$

The final model uses $\lambda_{\text{CKA}} : \lambda_{\text{CS}} = 16 : 1$, selected by the protocol in Sec. 4.3. This value gives greater weight to

GDSQ-VLA

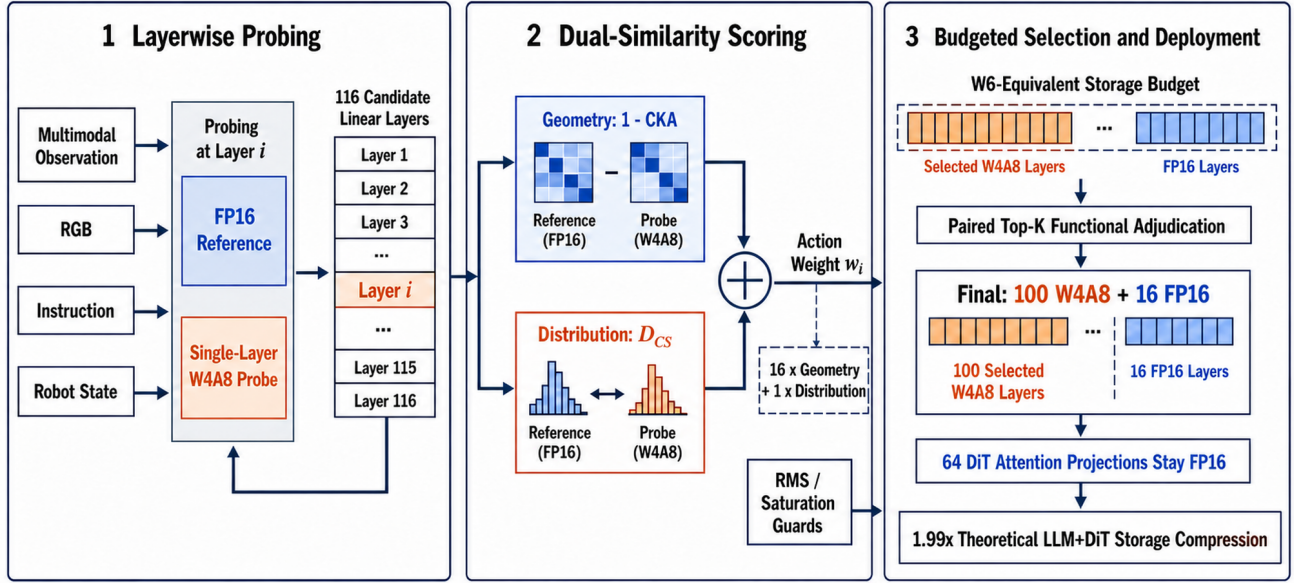


Figure 1. Overview of GDSQ-VLA. A single-layer intervention protocol measures complementary geometric and distributional representation changes, weights them by action-level sensitivity, and applies hard feasibility guards. A budgeted selector proposes diverse W4/FP16 masks, which are adjudicated by paired action-generation behavior before freezing the final 16:1 configuration. The 64 DiT attention projections outside the 116-layer candidate set remain in FP16.

the more stable geometry signal while retaining a nonzero distribution term; it is a checkpoint-level hyperparameter, not a universal constant.

3.3. Action-Weighted Constrained Selection

Representation distortion matters only insofar as it changes the action policy. We therefore estimate a layer importance d_i by applying the same isolated W4 intervention and comparing its action-generation trajectory with R under paired noise. The normalized action weight is

$$w_i = \frac{n d_i}{\sum_j d_j + \epsilon}, \quad (5)$$

followed by robust clipping and renormalization to keep $\frac{1}{n} \sum_i w_i = 1$. Thus, two layers with similar representation scores may receive different priorities when one perturbs the action solver more strongly.

Similarity alone cannot certify a usable activation range. For every intervention, we also measure a median log-RMS drift against the unwrapped FP16 output and estimate the fraction of values that would exceed the downstream static A8 range. The frozen thresholds are $\tau_{\text{RMS}} = 0.1319$ and $\tau_{\text{sat}} = 0.00232$, determined before final evaluation. A layer

violating either threshold cannot be assigned W4. These guards are engineering feasibility tests, not exact clipping measurements, because the saturation statistic approximates the downstream consumer with its calibrated A8 scale.

Let $x_i = 1$ assign W4A8 to candidate i and $x_i = 0$ leave its original FP16 Linear unchanged. We solve the binary allocation

$$\begin{aligned} \min_x \quad & \sum_i x_i w_i S_i, \\ \text{s.t.} \quad & \sum_i [x_i C_i^{\text{W4}} + (1 - x_i) C_i^{\text{FP16}}] \leq B, \\ & x_i = 0 \quad \text{if } G_i^{\text{RMS}} > \tau_{\text{RMS}} \text{ or } G_i^{\text{sat}} > \tau_{\text{sat}}, \\ & x_i \in \{0, 1\}. \end{aligned} \quad (6)$$

Here C_i is theoretical tightly packed weight storage and $B = 862,912,512$ bytes, the uniform-W6 reference budget for the candidate scope. The budget excludes activation peaks, CUDA workspaces, simulator memory, and fake-quant caches. Greedy allocation and diverse local bit flips generate a set of low-objective masks rather than treating the additive optimum as final.

3.4. Functional Adjudication and Final Freeze

Layer scores cannot model cancellation or amplification among simultaneous interventions. We therefore instantiate each diverse Top- K mask with its own static A8 calibration and compare the complete model with FP16 on paired observations and action noise. The functional score combines final-action error, kinematic error, gripper mismatch, and a tail penalty:

$$D_{\text{func}} = D_{\text{final}} + D_{\text{kin}} + D_{\text{grip}} + 2 \text{ CVaR}_{0.9}(D_{\text{solver}}). \quad (7)$$

The solver term follows the denoising trajectory rather than the environment trajectory. Its paired design reduces variance across masks, while the tail term prevents a good mean from hiding a few severe action failures.

Functional adjudication chooses the representative mask for each CKA:CS ratio. We then evaluate ratios $\{8, 16, 32, 64, 128, 256\} : 1$ and a CKA-only boundary on four predeclared RoboCasa365 atomic tasks, with 50 target scenarios per task. The primary rule is maximum task-macro success rate; lower D_{func} and then a larger ratio break exact ties. The winning 16:1 configuration reaches 80.5% and is frozen before evaluation on the other atomic tasks. Its final mask assigns W4A8 to 100 candidate layers and leaves 16 candidate layers in FP16. We do not apply ATM or OHB to the final method; their plan-specific static variant is retained only as an ablation.

4. Experiments

4.1. Experimental Setup

Benchmark and policy. We evaluate exclusively on RoboCasa365 v1.0.1 [24]. The completed frozen results use the corresponding GR00T N1.5 atomic-seen, composite-seen, and composite-unseen target post-training checkpoints at 60k steps; the unified results table also reserves rows for a planned $\pi_{0.5}$ study [27]. The benchmark provides long-horizon kitchen manipulation tasks with task-specific horizons and diverse initial scenes. Our atomic evaluation contains 18 tasks. Four tasks—OpenCabinet, OpenStandMixer-Head, PickPlaceDrawerToCounter, and CoffeeSetupMug—are declared as development tasks for selecting the CKA:CS ratio. The other 14 tasks form the primary evaluation set. We also report the 18-task average as a secondary, partially development-inclusive statistic.

Configurations. The FP16 policy is the uncompressed deployment reference and is excluded from best-compressed highlighting. For GR00T N1.5, the strongest low-bit baseline is the original QuantVLA W4A8 layout with checkpoint-specific static ATM and OHB [40]; it quantizes 116 of the 180 LLM+DiT Linear layers. GDSQ-VLA uses the frozen 16:1 mask and quantizes 100 layers. A fourth configuration applies plan-specific static ATM/OHB to the same

GDSQ-VLA mask, isolating whether scale correction remains useful after layer selection. All completed low-bit configurations use group size 64, 32 static-A8 calibration batches, P99.9 activation scales, and the same frozen calibration buffer. The $\pi_{0.5}$ configurations remain predeclared placeholders until their model-specific masks and evaluations are complete.

Rollout protocol. Each configuration is evaluated on 50 scenarios per task, giving 700 episodes for the 14-task atomic primary set, 900 for all 18 atomic tasks, and 800 for each 16-task composite group. We use official task horizons, four denoising steps, and execute 16 actions per replan. To obtain paired comparisons, every configuration receives a deterministic standard-normal initial action noise keyed by (task, environment seed, replan index). This common-random-number design preserves the marginal noise distribution of each policy while reducing variance in configuration differences. It is a paired evaluation protocol and should not be conflated with an implementation that draws native GPU random noise independently for each configuration.

Statistics. Success rate (SR) is first averaged within each task and then macro-averaged across tasks. We report 10,000-draw task-cluster bootstrap confidence intervals, paired task-level sign-flip tests, and Holm correction across the five predeclared contrasts. This treats tasks, rather than individual frames or replans, as the unit of generalization. The incomplete $\pi_{0.5}$ evaluation appears only as **TBD** in the unified table; running partial logs are never reported.

4.2. RoboCasa365 Results

Table 1 keeps the three official RoboCasa365 task groups in one view. On the completed GR00T N1.5 Atomic evaluation, GDSQ-VLA reaches 69.0% over all 18 tasks, compared with 49.6% for QuantVLA W4A8+ATM/OHB and 75.6% for the uncompressed reference. This aggregate includes the four development tasks and is therefore a benchmark overview rather than the selection-controlled primary result.

The completed Composite-Seen split tests whether the atomic-selected mask transfers to longer, multi-stage tasks without further tuning. GDSQ-VLA reaches 40.6% SR, compared with 19.6% for QuantVLA W4A8+ATM/OHB and 41.9% for FP16. Its paired gain over the low-bit baseline is 21.0 points with a 95% task-cluster interval of [14.3, 28.0] and Holm-adjusted $p = 0.0003$; its -1.25 -point difference from FP16 has interval $[-5.75, 3.00]$. This split supports transfer to seen composite tasks; the separate frozen split below tests unseen compositions.

The frozen Composite-Unseen split evaluates 16 unseen task compositions without changing the atomic-selected ratio or layer budget. GDSQ-VLA reaches 40.4% SR, compared with 19.8% for QuantVLA W4A8+ATM/OHB and

Table 1. Unified RoboCasa365 results across the official Atomic, Composite-Seen, and Composite-Unseen task groups. Atomic reports the All-18 macro success rate; the selection-controlled Primary-14 result is reported separately in Table 3. Green cells mark the best compressed configuration within the same policy, excluding the uncompressed FP16 reference. Deployment storage is a theoretical tightly packed Linear-layer estimate, not live CUDA memory. **TBD** denotes a predeclared evaluation that has not been completed and frozen; no partial logs are used.

Configuration	W4A8 layers	Atomic	Composite-Seen	Composite-Unseen	Deployment	
		All 18 SR (%) ↑	SR (%) ↑	SR (%) ↑	Storage (GiB) ↓	Compression ↑
<i>GR00T N1.5 — 50 paired scenarios per task</i>						
FP16 (uncompressed)	0	75.6	41.9	45.3	1.993	Uncompressed
QUANTVLA W4A8 + ATM/OHB	116	49.6	19.6	19.8	0.898	2.22×
GDSQ-VLA + ATM/OHB	100	66.7	39.1	40.3	1.001	1.99×
GDSQ-VLA (Ours)	100	69.0	40.6	40.4	1.001	1.99×
<i>$\pi_{0.5}$ — planned RoboCasa365 evaluation</i>						
FP16 (uncompressed)	0	TBD	TBD	TBD	TBD	Uncompressed
QUANTVLA W4A8 + ATM/OHB	TBD	TBD	TBD	TBD	TBD	TBD
GDSQ-VLA (Ours)	TBD	TBD	TBD	TBD	TBD	TBD

45.3% for FP16. Its paired gain over the low-bit baseline is 20.6 points with interval $[14.5, 26.8]$ and Holm-adjusted $p = 0.0002$. The remaining -4.88 -point gap to FP16 has interval $[-8.12, -1.75]$ and adjusted $p = 0.0316$. Thus, the allocation transfers substantially better than the fixed W4A8 layout to unseen compositions, although it is not lossless relative to full precision.

Across all 50 official tasks and 2,500 episodes per configuration, GDSQ-VLA obtains 50.8% macro SR, compared with 30.4% for QuantVLA W4A8+ATM/OHB and 55.1% for FP16. The corresponding paired differences are $+20.3$ points with interval $[16.9, 24.0]$ and -4.32 points with interval $[-6.56, -2.08]$, respectively. Static ATM/OHB changes the 50-task result from 50.8% to 49.4%; its -1.36 -point interval $[-3.28, 0.52]$ spans zero, so we retain the simpler selected mask as the final configuration.

The final model stores an estimated 1.001 GiB over the LLM+DiT Linear scope, compared with 1.993 GiB for FP16, corresponding to 49.8% savings or $1.99\times$ compression. The estimate assumes tightly packed W4 weights, FP16 per-output scales, FP32 block rotations, and FP16 for unselected layers and biases. It excludes vision parameters, activation peaks, CUDA workspaces, simulator memory, and fake-quant caches. This accounting matches the component scope used for the baseline comparison but is not a measurement of current process memory.

The same table reserves $\pi_{0.5}$ cells so that the planned cross-policy study remains visible without mixing incomplete evidence into the GR00T N1.5 results. These entries stay **TBD** until each complete matrix is frozen and validated. We make no cross-policy transfer claim from placeholders.

4.3. Ratio Selection and Held-Out Evaluation

Table 2 demonstrates why the representation proxy is a search guide rather than the final objective. The 64:1 config-

Table 2. Predeclared ratio selection on four development tasks (50 scenarios per task). Lower D_{func} is better, but the final ratio is selected by closed-loop macro success rate.

CKA:CS ratio	$D_{\text{func}} \downarrow$	Dev SR (%) ↑
8:1	0.11597	77.0
16:1	0.10044	80.5
32:1	0.13395	76.5
64:1	0.07495	75.0
128:1	0.10236	77.0
256:1	0.09328	79.5
CKA only	0.09157	77.5

Table 3. Selection-controlled GR00T N1.5 Atomic results. The CKA:CS ratio is frozen using four development tasks; none of the Primary 14 tasks contributes to ratio selection. Confidence intervals are 95% task-cluster bootstrap intervals. Green marks the best compressed configuration, excluding FP16.

Configuration	Primary 14 SR (%) ↑
FP16 (uncompressed)	73.9 [66.0, 80.9]
QUANTVLA W4A8 + ATM/OHB	45.6 [34.4, 57.0]
GDSQ-VLA + ATM/OHB	63.0 [52.0, 72.9]
GDSQ-VLA (Ours)	65.9 [57.3, 73.7]

uration achieves the lowest D_{func} on its paired calibration observations, but only 75.0% closed-loop development SR. The 16:1 configuration instead achieves the highest development SR at 80.5% (161/200 scenarios) and is frozen by the predeclared rule. CKA-only reaches 77.5%, indicating that the CS term should be down-weighted but not removed for this checkpoint. The non-monotonic sweep also argues against inferring a universal ratio from proxy correlation alone.

Table 3 reports the selection-controlled result after the

Table 4. Five-seed diagnostic ablation. This screen predates the 50-scenario ratio freeze and therefore studies signal composition rather than the final 16:1 mask. Green cells compare quantized configurations only.

Configuration	Held-out 14 SR	All 18 SR
FP16 (uncompressed)	71.4	73.3
W4A8 + ATM/OHB	52.9	52.2
CS only	58.6	61.1
CKA only	57.1	60.0
CKA + ATM/OHB	60.0	63.3
CS+CKA (64:1)	67.1	67.8

16:1 ratio is frozen. On the 14 tasks excluded from ratio selection, GDSQ-VLA reaches 65.9% SR, whereas the original QuantVLA W4A8+ATM/OHB recipe reaches 45.6%. The paired gain is 20.3 percentage points with a 95% task-cluster interval of [13.9, 26.9]; the exact sign-flip test remains significant after Holm correction ($p = 0.0012$).

Layer selection does not close the full-precision gap. FP16 obtains 73.9% on the Primary 14 tasks, exceeding GDSQ-VLA by 8.0 points with a paired interval of $[-12.1, -3.9]$ for GDSQ-VLA minus FP16 and Holm-adjusted $p = 0.0073$. The result therefore supports a precision-allocation improvement over the fixed low-bit layout, not lossless quantization. The final mask trades 7.8 percentage points of theoretical storage savings for 20.3 points of task success relative to the W4A8 baseline.

4.4. Component and Calibration Ablations

The five-seed diagnostic in Table 4 separates the two representation signals under a smaller screening protocol. CS-only and CKA-only reach 58.6% and 57.1% on the 14 held-out tasks, respectively, while a 64:1 combination reaches 67.1%. The task-cluster intervals in this screen overlap, so we treat the result as evidence that the signals can be complementary, not as a significance claim for the 64:1 mask. The higher-powered ratio study subsequently replaces 64:1 with the final 16:1 configuration.

Static ATM/OHB does not improve the selected final mask. In the 50-scenario evaluation, adding it changes primary SR from 65.9% to 63.0%, a paired difference of -2.9 points with interval $[-6.9, 0.7]$ and adjusted $p = 0.226$. This negative, non-significant point estimate motivates leaving ATM/OHB out of GDSQ-VLA. The result also illustrates that a calibration designed for a fixed quantization layout need not remain helpful after sensitive layers have already been protected in FP16.

4.5. Runtime and Deployment Scope

Table 5 exposes the current implementation boundary. The eager W4A8 path retains floating-point weights and caches,

Table 5. Observed eager fake-quant runtime. These measurements explain why we claim theoretical component storage rather than end-to-end speed or live-memory gains. Green cells compare compressed configurations only; FP16 is an uncompressed reference.

Configuration	Wall (s)	Replan (s)	Peak MiB
FP16 (uncompressed)	26.9	0.139	12,769
QuantVLA+ATM/OHB	37.5	0.268	17,973
GDSQ-VLA +ATM/OHB	31.6	0.259	17,719
GDSQ-VLA (Ours)	30.9	0.254	17,671

applies block transformations, and ultimately executes floating-point GEMMs. Consequently, GDSQ-VLA requires 0.254 seconds per replan and peaks at 17.7 GiB, compared with 0.139 seconds and 12.8 GiB for FP16. Episode wall time also depends on how early successful tasks terminate and is not a pure model benchmark. Realizing the theoretical storage benefit in live memory and latency requires fused INT4/INT8 kernels and removal of redundant floating-point buffers; those systems changes are orthogonal to the layer-allocation policy evaluated here.

5. Conclusion and Limitations

This paper introduced GDSQ-VLA, a training-free W4/FP16 layer-selection framework for quantized VLA policies. Its central idea is to treat quantization sensitivity as a combination of representation geometry, distribution overlap, and downstream action behavior. CKA and CS divergence guide layer-level search, action sensitivity weights the local evidence, hard guards reject unstable interventions, and paired functional adjudication selects complete masks. On 14 RoboCasa365 atomic tasks excluded from ratio selection, this design improves success by 20.3 points over the original QuantVLA W4A8+ATM/OHB recipe. The same 20.3-point gain holds across all 50 official tasks, while the final mask retains a theoretical $1.99\times$ LLM+DiT component-storage compression.

The present evidence has three boundaries. First, the frozen allocation is evaluated on one GR00T N1.5 check-point family and a binary W4/FP16 search space; the planned $\pi_{0.5}$ rows remain TBD, so broader model transfer is not yet supported. Second, GDSQ-VLA remains 8.0 success points below FP16 on the primary atomic tasks and 4.9 points below FP16 on Composite-Unseen, despite recovering 20.6 points over the fixed low-bit baseline on that unseen split. Across all 50 tasks, the remaining FP16 gap is 4.3 points. Third, the eager fake-quant runtime is slower and uses more live GPU memory than FP16, so our deployment claim is limited to theoretical tightly packed storage. Future work should validate the selection rule on additional VLA architectures and pair it with fused low-bit kernels that realize the predicted storage benefit in an end-to-end system.

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